

GenAI Container Challenge (15–20 mins)

Problem:

Build and ship a production-ready GenAI API container (prompt → response).

Tasks

1. App

- Create API: POST /generate
- Input: prompt
- Output: response (mock or real LLM)

2. Docker

- Multi-stage Dockerfile
- Slim/minimal base image
- Optimize caching
- Build using BuildKit / Buildx

3. Security

- Generate SBOM
- Scan using docker scout / trivy
- Fix at least 2 issues

4. Hardening

- Run as non-root
- Add HEALTHCHECK
- Minimize attack surface

5. Dev Environment

- Create .devcontainer/devcontainer.json
- Include docker sbom, docker scout, docker ai

Deliverables

- App code
- Dockerfile
- Build command

- SBOM + scan result
- Secure container
- DevContainer config

Bonus

- Add real LLM
- Multi-arch build
- Use docker ai